TrES-1b

R-1766

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_190902` · created 2026-08-02T21:34:37+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458728.8086 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.014
Transit depth (Rp/Rs)^2	2.52 +/- 0.43 [%]
Orbital Inclination (inc)	88.0 +/- 0.87
Airmass coefficient 1 (a1)	0.951 +/- 0.015
Airmass coefficient 2 (a2)	0.026 +/- 0.0042
Scatter in the residuals of the lightcurve fit is	1.55 %
Best Comparison Star	#1 - [414, 115]
Optimal Aperture	4.87
Optimal Annulus	15.75
Transit Duration (day)	0.0997 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.014 vs 0.1358 (deviation 1.66σ)
Duration fit vs published	0.0997 d vs 0.1042 d (deviation 0.8σ)
Mid-time O–C	5.3 min
Depth SNR	11.4
Residual scatter	1.55 %

Light curve

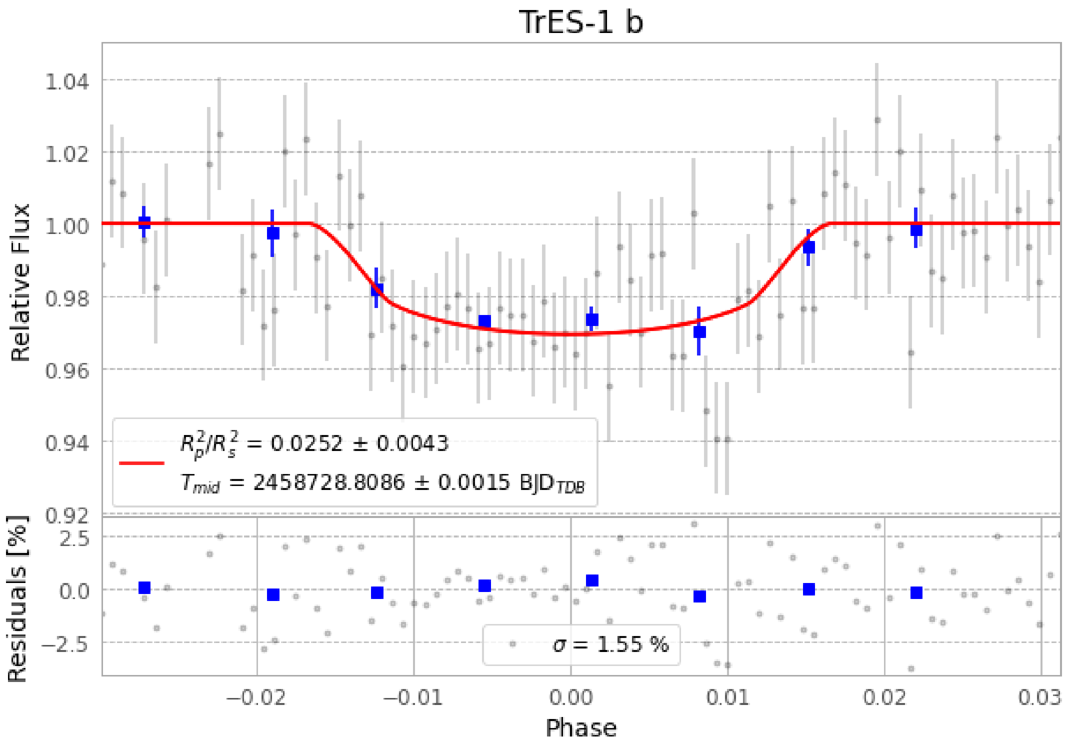

Figure 1 — FinalLightCurve_TrES-1 b_2019-09-01.png (SHA-256 8d341bf60c3660b1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 249], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 0c172f6d0fcb5d89…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

90 FITS frames (59 MB), `TRES-1190902050913.FITS` → `TRES-1190902093612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-190902.log (2f11c15dd959…), FinalLightCurve_TrES-1 b_2019-09-01.png (8d341bf60c36…), FinalLightCurve_TrES-1 b_2019-09-01.csv (97b00ff992f0…)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)